

ELGIN 5 S, AZ, USA

WMO: 722760

Lat: 31.5907N

Lon: 110.5087W

Elev: 1466

StdP: 84.91

Time Zone: -7.00 (W07)

Period: 02-19

WBAN: 53132

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.3	-4.5	-19.6	0.8	17.9	-17.3	1.0	15.2	8.0	9.8	6.8	9.3	0.9	N/A	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.8	35.1	14.8	33.8	15.1	32.5	15.1	19.7	27.0	19.2	26.4	18.8	25.9	3.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.4	21.2	17.4	14.9	21.0	16.8	14.4	20.9	63.2	27.0	61.5	26.4	59.9	25.7	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.7	5.8	5.1	DB	-10.5	37.1	2.4	1.2	-12.2	38.0	-13.6	38.7	-14.9	39.4	-16.7	40.3	(4)
(5)			WB	-12.2	20.6	2.7	0.5	-14.1	21.0	-15.7	21.3	-17.2	21.6	-19.1	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	7.1	8.3	11.5	14.4	18.4	24.1	24.5	23.1	21.1	16.5	11.3	7.1	(6)	
	DBStd	7.03	3.42	3.90	3.62	3.32	3.30	2.24	2.00	1.70	1.99	3.06	3.62	3.70	(7)	
	HDD10.0	335	98	71	27	7	1	0	0	0	0	1	28	104	(8)	
	HDD18.3	1634	347	282	211	122	40	0	0	0	3	70	211	348	(9)	
	CDD10.0	2398	9	22	74	137	262	422	448	407	332	204	67	14	(10)	
	CDD18.3	656	0	0	0	3	42	172	190	149	86	14	0	0	(11)	
	CDH23.3	8564	0	14	78	252	993	2604	1841	1334	998	404	46	1	(12)	
	CDH26.7	3602	0	1	5	37	352	1402	862	521	342	78	2	0	(13)	
(14) Wind	WSAvg	2.2	2.0	2.3	2.5	2.7	2.7	2.5	2.1	1.7	1.7	1.9	2.0	2.0	(14)	
(15) Precipitation	PrecAvg	439	33	28	22	10	5	14	107	99	41	19	20	33	(15)	
	PrecMax	608	177	96	91	45	34	67	203	161	94	189	83	104	(16)	
	PrecMin	259	0	0	0	0	0	0	33	46	14	0	0	0	(17)	
	PrecStd	104	42	23	22	14	8	16	46	28	19	39	19	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.8	25.3	27.3	29.5	33.5	37.2	36.1	34.3	32.9	30.6	26.7	22.9	(19)	
		MCWB	7.8	8.8	9.9	11.1	12.7	14.8	16.1	17.8	16.2	13.2	10.0	8.3	(20)	
	2%	DB	19.9	22.9	25.5	27.7	31.7	35.7	34.6	32.5	31.4	28.7	24.7	20.6	(21)	
		MCWB	7.3	8.0	9.2	10.5	12.1	14.4	16.6	17.7	16.0	12.9	10.1	7.7	(22)	
	5%	DB	18.2	20.5	23.8	26.3	30.3	34.5	33.1	31.2	30.2	27.3	22.9	18.3	(23)	
		MCWB	6.6	7.4	8.6	9.9	11.5	14.1	16.8	17.8	16.0	12.7	9.7	7.5	(24)	
	10%	DB	16.2	17.9	22.0	24.7	28.8	33.2	31.5	29.9	28.9	25.8	21.0	15.8	(25)	
		MCWB	5.8	6.6	8.0	9.4	10.9	14.0	17.0	17.7	15.9	12.3	9.0	6.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.7	10.8	12.4	13.7	18.3	20.2	20.7	19.5	16.8	13.0	11.0	(27)	
		MCDB	18.1	16.1	24.3	27.0	30.5	27.1	27.4	28.3	25.9	24.1	20.6	15.2	(28)	
	2%	WB	8.8	9.7	10.0	11.4	12.8	17.4	19.6	20.0	18.8	15.6	11.4	9.5	(29)	
		MCDB	16.4	17.8	23.5	26.0	29.7	28.0	26.5	27.6	25.4	24.3	19.6	17.3	(30)	
	5%	WB	7.8	8.7	9.3	10.6	11.9	16.7	19.1	19.5	18.2	14.7	10.6	8.5	(31)	
		MCDB	16.1	18.4	22.3	24.9	28.3	28.0	26.0	27.0	24.9	23.6	19.8	16.8	(32)	
	10%	WB	6.7	7.7	8.6	9.9	11.1	15.9	18.7	19.0	17.7	13.6	9.8	7.4	(33)	
		MCDB	15.0	17.6	20.9	23.9	26.9	28.3	25.6	26.3	24.5	22.5	19.3	15.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.2	16.5	17.6	18.0	18.2	17.8	13.1	13.0	14.8	17.1	17.1	15.7	(35)	
		MCDBR	19.6	21.1	20.8	20.8	20.1	19.4	15.8	15.5	17.4	19.0	19.8	19.1	(36)	
	5% WB	MCWBR	9.8	9.8	9.3	9.1	8.3	7.1	4.3	4.6	5.9	7.8	8.9	9.4	(37)	
		MCWBR	16.7	17.7	19.5	18.8	19.2	15.8	12.1	12.5	12.9	16.3	16.4	16.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.244	0.247	0.264	0.272	0.286	0.311	0.363	0.351	0.313	0.279	0.254	0.243	(40)	
		taud	2.683	2.676	2.597	2.558	2.555	2.487	2.463	2.488	2.572	2.620	2.653	2.675	(41)	
	Ebn at Noon		1011	1032	1027	1020	999	967	916	927	958	979	989	996	(42)	
		Edh at Noon	72	80	93	101	102	109	112	107	95	83	73	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.71	4.65	6.13	7.50	8.21	8.01	6.41	6.24	5.82	5.11	4.09	3.41	(44)	
	RadStd		0.25	0.37	0.34	0.26	0.32	0.47	0.47	0.25	0.29	0.32	0.20	0.15	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-75	N/A	N/A

Nomenclature: See separate page